Solutions: Midterm Exam 2

ECON 300 – Labour Economics

Answers and Explanations

1. **B**
   A monopolist uses MRP, which lies below VMP and is therefore less elastic.

2. **A**

Economic Intuition

Economic rent is defined as the difference between the wage a worker receives and the worker's **reservation wage** (the minimum wage required to induce them to work).

$$\text{Economic Rent} = W - W^R$$

where:

- W = actual wage
- W^R = reservation wage

If workers are earning economic rents, then:

- They are being paid **more than necessary** to keep them employed.
- A reduction in wages **will not immediately reduce employment**.
- Workers will continue working as long as:

$$W \geq W^R$$

Why Answer A is Correct

Answer A reflects the idea that:

> A firm can reduce wages **without losing workers**, as long as wages remain above the reservation wage.

This is the defining property of economic rents:

- Firms have some flexibility to lower wages.
- Workers do not leave immediately because they are still better off than their next best alternative.

Why Answer D is Incorrect

Answer D states that workers would leave if wages are reduced. This is incorrect because:

- Workers only leave when:
$$W < W^R$$
- A small wage reduction **within the rent margin** does not trigger exit.

Thus, Answer D ignores the key concept of a **reservation wage threshold**.

Conclusion

$$\boxed{\text{Economic rents imply that wages can fall without reducing employment, as long as } W \geq W^R.}$$

Therefore, **Answer A is correct**.

3. **B**
   Monopsony implies upward-sloping supply, so $MCL > W$.

4. **A**
   Industry supply is upward sloping as higher wages attract more workers.

5. **A**
   Payroll taxes increase labour costs and reduce employment.

6. **C**
   Tax burden is shared depending on elasticities.

7. **D**
   Elasticity $= \frac{\%\Delta N}{\%\Delta W}$.

8. **E**
   Isoprofit curves are concave due to diminishing trade-offs between safety and wages.
9. **D**
   Efficiency wages relate to productivity channels, not directly safety.
10. **B**
    Indifference curves reflect diminishing marginal rate of substitution.
11. **A**
    Risk-averse workers require higher compensation for risk.
12. **A**
    Higher isoprofit curves imply lower profits.
13. **B**
    Slope of wage-risk locus reflects compensating differential.
14. **E**
    Adam Smith introduced compensating wage differentials.
15. **B**
    Workers respond to incentives; statement denying this is false.
16. **D**
    Competitive equilibrium assumes perfect information.
17. **A**
    Risk-averse workers choose safer jobs with lower wages.
18. **E**
    Uber drivers accept lower wages for flexibility.
19. **B**
    Compensation is for job characteristics, not utility itself.
20. **D**
    Education signals ability rather than directly raising productivity.
21. **B**
    Human capital increases productivity and earnings.
22. **C**
    Opportunity cost is foregone earnings.
23. **C**
    Private costs/benefits accrue to individuals.
24. **D**
    We are looking at the earning profiles in present value terms. There is no difference between old and young individual's PV of earnings.

25. **E**
Perfect capital markets remove current income constraints.

26. **A**
Optimal schooling occurs where discounted MB = MC.

27. **A**
Unknown variables imply imperfect information.

28. **C**
Earnings rise until mid-career then decline.

29. **A**
$PV = \frac{Y}{r}$ approximates long streams.

30. **B**
Monopsony outcome: $MCL = VMP$, giving (W_1, N_1).

31. **C**
Competitive equilibrium: supply equals demand (W_C, N_C).

32. **A**

In a monopsony labour market, the firm faces an upward-sloping labour supply curve. As a result:

- The firm chooses employment where:
$$MCL = VMP$$
- This determines employment N_M.
- The wage is then determined from the labour supply curve at that level of employment, giving wage W_M.

Vacancies arise when there is **excess demand for labour**:

$$\text{Vacancies} = N^d - N^s$$

In a monopsony:

- The firm sets both wage and employment optimally.
- It hires exactly the number of workers it wants at the chosen wage.
- Workers are willing to supply exactly that amount of labour at W_M.

Therefore:

$$N^d = N^s \quad \text{at the monopsony equilibrium}$$

Even though the monopsony equilibrium is **inefficient** relative to the competitive outcome:

- It is still an **equilibrium** from the firm's perspective.
- There is **no excess demand or excess supply**.
- Hence, there are **no vacancies**.

Conclusion

$$\boxed{\text{The number of vacancies in a monopsony equilibrium is zero.}}$$

Thus, the correct answer is:

$$\boxed{\text{A}}$$

33. **A**
    Area d represents direct costs (tuition, books).

34. **C**
    Area e represents lifetime earnings gain from high school diploma vs high school drop-out.

35. **E**
    Areas $b + c + d$ represent total cost of education.

Final Answer Summary

1. B	2. A	3. B	4. A	5. A	6. C
7. D	8. E	9. D	10. B	11. A	12. A
13. B	14. E	15. B	16. D	17. A	18. E
19. B	20. D	21. B	22. C	23. C	24. D
25. E	26. A	27. A	28. C	29. A	30. B
31. C	32. A	33. A	34. C	35. E	

Solution Key: Human Capital Investment Decision

Given Information

Johny is considering a **3-year college diploma**.

- Length of program: 3 years
- Annual tuition: \$1,200
- Annual books/materials: \$300

- Earnings without college: \$8,000 per year
- Earnings with diploma: \$14,000 per year
- Working life after graduation: 12 years
- Market interest rate: $r = 6\% = 0.06$

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1. Total annual direct cost of attending college

The annual **direct cost** is:

$$\text{Direct Cost} = \text{Tuition} + \text{Books}$$

$$\text{Direct Cost} = 1200 + 300 = 1500$$

$$\boxed{\text{Annual direct cost} = \$1{,}500}$$

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2. Opportunity cost per year of attending college

The **opportunity cost** is the income Johny gives up by going to school instead of working.

$$\text{Opportunity Cost per year} = \$8{,}000$$

$$\boxed{\text{Opportunity cost per year} = \$8{,}000}$$

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3. Total cost of the investment over the 3-year period (ignoring discounting)

The total annual cost is:

$$\text{Annual Total Cost} = \text{Direct Cost} + \text{Opportunity Cost}$$

$$\text{Annual Total Cost} = 1500 + 8000 = 9500$$

Since the program lasts 3 years:

$$\text{Total Cost over 3 years} = 3 \times 9500 = 28{,}500$$

$$\boxed{\text{Total undiscounted cost over 3 years} = \$28{,}500}$$

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4. Annual earnings gain from completing the diploma

The additional annual earnings from the diploma are:

$$\Delta Y = 14{,}000 - 8{,}000 = 6{,}000$$

$$\boxed{\text{Annual earnings gain} = \$6{,}000}$$

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5. Present value of benefits

Johny receives the additional earnings of \$6,000 each year for 12 years after graduation.

So the present value of benefits, measured **at the time of graduation**, is the present value of a 12-year annuity:

$$PV_B^{(t=3)} = 6000 \left(\frac{1 - (1.06)^{-12}}{0.06} \right)$$

First compute:

$$(1.06)^{12} \approx 2.0122$$

$$(1.06)^{-12} \approx \frac{1}{2.0122} \approx 0.49697$$

Then:

$$\frac{1 - 0.49697}{0.06} = \frac{0.50303}{0.06} \approx 8.3838$$

So:

$$PV_B^{(t=3)} = 6000(8.3838) \approx 50{,}302.8$$

This is the present value **at the time Johny finishes school**. Since the schooling decision is made **today**, we must discount this amount back 3 years:

$$PV_B^{(t=0)} = \frac{50{,}302.8}{(1.06)^3}$$

$$(1.06)^3 = 1.191016$$

$$PV_B^{(t=0)} \approx \frac{50{,}302.8}{1.191016} \approx 42{,}235.9$$

$$\boxed{PV(\text{Benefits}) \approx \$42{,}236}$$

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6. Present value of costs

Johny incurs total annual cost of \$9,500 for 3 years. These costs occur during years 1, 2, and 3.

Thus the present value of costs is:

$$PV_C = \frac{9500}{1.06} + \frac{9500}{(1.06)^2} + \frac{9500}{(1.06)^3}$$

Compute each term:

$$\frac{9500}{1.06} \approx 8962.26$$

$$\frac{9500}{1.1236} \approx 8454.97$$

$$\frac{9500}{1.191016} \approx 7974.71$$

Add them:

$$PV_C \approx 8962.26 + 8454.97 + 7974.71 = 25{,}391.94$$

$$\boxed{PV(\text{Costs}) \approx \$25{,}392}$$

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7. Net Present Value (NPV) decision

The net present value is:

$$NPV = PV(\text{Benefits}) - PV(\text{Costs})$$

$$NPV = 42{,}235.9 - 25{,}391.94 = 16{,}843.96$$

$$\boxed{NPV \approx \$16{,}844}$$

Since:

$$NPV > 0$$

Johny **should invest** in the diploma.

$$\boxed{\text{Yes, Johny should undertake the investment.}}$$

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8. Effect of a higher interest rate ($r = 8\%$)

If the market interest rate rises, future earnings are discounted more heavily.
This means:

- The present value of future benefits falls.
- The attractiveness of education decreases.

Now compute the present value of benefits at 8%:

$$PV_B^{(t=3)} = 6000\left(\frac{1-(1.08)^{-12}}{0.08}\right)$$

$$(1.08)^{12} \approx 2.5182$$

$$(1.08)^{-12} \approx 0.3971$$

$$\frac{1-0.3971}{0.08} = \frac{0.6029}{0.08} \approx 7.5363$$

$$PV_B^{(t=3)} \approx 6000(7.5363) = 45{,}217.8$$

Discounting back 3 years:

$$PV_B^{(t=0)} = \frac{45{,}217.8}{(1.08)^3}$$

$$(1.08)^3 = 1.2597$$

$$PV_B^{(t=0)} \approx \frac{45{,}217.8}{1.2597} \approx 35{,}895.7$$

Now compute the present value of costs:

$$PV_C = \frac{9500}{1.08} + \frac{9500}{(1.08)^2} + \frac{9500}{(1.08)^3}$$

$$\frac{9500}{1.08} \approx 8796.30$$

$$\frac{9500}{1.1664} \approx 8144.15$$

$$\frac{9500}{1.2597} \approx 7541.48$$

$$PV_C \approx 8796.30 + 8144.15 + 7541.48 = 24{,}481.93$$

Thus:

$$NPV_{8\%} \approx 35{,}895.7 - 24{,}481.93 = 11{,}413.77$$

So the NPV is still positive, but smaller.

$$\boxed{\text{A higher interest rate reduces the attractiveness of education, but Johny should still invest.}}$$

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9. Approximate internal rate of return (IRR)

The IRR is the discount rate that makes:

$$PV(\text{Benefits}) = PV(\text{Costs})$$

We already know:

- At 6%, NPV is positive.
- At 8%, NPV is still positive.

So the IRR must be **above 8%**.

Try 12%:

$$PV_B^{(t=3)} = 6000\left(\frac{1-(1.12)^{-12}}{0.12}\right)$$

$$(1.12)^{12} \approx 3.89598$$

$$(1.12)^{-12} \approx 0.2567$$

$$\frac{1-0.2567}{0.12} = \frac{0.7433}{0.12} \approx 6.194$$

$$PV_B^{(t=3)} \approx 6000(6.194) = 37{,}164$$

Discount back 3 years:

$$PV_B^{(t=0)} = \frac{37{,}164}{(1.12)^3}$$

$$(1.12)^3 = 1.404928$$

$$PV_B^{(t=0)} \approx 26{,}453$$

Now costs at 12%:

$$PV_C = \frac{9500}{1.12} + \frac{9500}{(1.12)^2} + \frac{9500}{(1.12)^3}$$

$$\approx 8482.14 + 7573.34 + 6761.91 = 22{,}817.39$$

So:

$$NPV_{12\%} \approx 26{,}453 - 22{,}817 = 3{,}636$$

Still positive, so IRR is above 12%.

Try 15%:

Benefits:

$$PV_B^{(t=3)} = 6000\left(\frac{1-(1.15)^{-12}}{0.15}\right)$$

$$(1.15)^{12} \approx 5.3503, \qquad (1.15)^{-12} \approx 0.1869$$

$$\frac{1-0.1869}{0.15} \approx 5.4207$$

$$PV_B^{(t=3)} \approx 6000(5.4207) = 32{,}524.2$$

Discount back 3 years:

$$(1.15)^3 = 1.520875$$

$$PV_B^{(t=0)} \approx \frac{32{,}524.2}{1.520875} \approx 21{,}385$$

Costs:

$$PV_C = \frac{9500}{1.15} + \frac{9500}{(1.15)^2} + \frac{9500}{(1.15)^3}$$

$$\approx 8260.87 + 7183.37 + 6246.41 = 21{,}690.65$$

Now NPV is slightly negative:

$$NPV_{15\%} \approx 21{,}385 - 21{,}691 \approx -306$$

So the IRR is between 14% and 15%, very close to 15%.

$$\boxed{IRR \approx 14.8\% \text{ to } 15\%}$$

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10. Comparative statics

(a) If expected post-education earnings fall

Then:

$$\Delta Y = \text{earnings with diploma} - \text{earnings without diploma}$$

becomes smaller.
Therefore:

- present value of benefits falls,
- NPV falls,
- IRR falls.

So Johny becomes **less likely** to invest.

$$\boxed{\text{Lower post-education earnings reduce the attractiveness of schooling.}}$$

(b) If tuition fees increase

Higher tuition raises direct cost D, so:

- marginal cost rises,
- present value of costs rises,
- NPV falls.

Thus Johny becomes **less likely** to invest.

$$\boxed{\text{Higher tuition reduces the incentive to invest in education.}}$$

(c) If his working life becomes shorter

A shorter working life means fewer years over which to earn the higher post-education income.
Thus:

- present value of benefits falls,
- NPV falls,

- IRR falls.

So Johny becomes **less likely** to invest.

$$\boxed{\text{A shorter working life reduces the return to education.}}$$

Final Conclusion

Johny should invest in the 3-year diploma because:

$$PV(\text{Benefits}) > PV(\text{Costs})$$

and

$$NPV > 0$$

The estimated internal rate of return is approximately:

$$\boxed{IRR \approx 15\%}$$

which is well above the market interest rate of 6%.

$$\boxed{\text{Therefore, the diploma is a worthwhile human capital investment.}}$$